Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	AllStar
Project / Solution Name	Orbyte Smart SME
Name (Leader)	PHAN YI YAN
Email & Phone (Leader)	phanyiyan@gmail.com 011-12104025
University Affiliation(s)	Universiti Sains Malaysia (USM)
Live Prototype URL	https://thesaurus-tracing-faceplate.ngrok-free.dev/#overview
Code Repository URL (public / access granted to judges)	https://github.com/WXY0609/Orbyte-Smart-SME-Challenge.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The system follows a modern full-stack web application architecture. The frontend is built using React (Vite) styled with Tailwind CSS and Lucide React icons to make an interactive SME operations and waste management dashboard. Frontend user interactions and queries are sent to a Python Flask API backend via RESTful HTTP requests. The Flask backend processes operational logic, data collection and handles secure API integrations with generative AI endpoints (Google Gemini API). The AI model analyses operational data to generate real-time optimization insights, waste reduction suggestions and automated decision support. All processed metrics are returned to the frontend dashboard for real-time visualization.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.10+	Backend server runtime & core API application logic	PSF License
Flask (v3.x)	API backend framework	BSD-3-Clause
JavaScript / Node.js	Frontend development and application host	MIT License
React (v18.x)	Frontend user interface framework	MIT License
Vite (v5.x)	Frontend build tool and local development server	MIT License
Tailwind CSS (v3.x)	Utility-first CSS framework for dashboard styling	MIT License
Lucide React	UI icon component library	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (Google)	Generative AI for waste management optimization, SME operational analytics and decision support	Free Tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Local Host / Render / Ngrok	Hosting frontend (React) and backend (Flask) for prototype demonstration	Free / Developer Tier

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 1.5 Flash / Pro	Google DeepMind	Pre-trained	Google AI Terms of Service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
SME Waste & Operations Sample Dataset	Self-collected	Synthetic	Open	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Gemini	Assisting with code generation, UI layout debugging, and documentation drafting	major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-assisted code generation, UI designs, and documentation drafts were thoroughly reviewed, refined and tested by team members before integration. We performed integration tests across the React frontend components and Flask REST API endpoints to ensure code reliability, data safety and full compatibility with project requirements. The team takes full responsibility for the final implementation.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☑ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
None	None	None

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

None

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No user data is collected

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Technical limitations include AI API rate limits, model response speed and risk of inaccurate AI generation. Mitigations implemented include front end validation, structured prompt engineering to control AI outputs to predictable formats, error handling for API timeouts and fallback UI states to ensure smooth user experience during network delays.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



PHAN YI YAN (Group Leader)
25/10/2026